

On the sum of binomial coefficients indexed by the prime divisors of n

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1 The $f(n)$ Function

This is a study into interesting properties and relationships generated by the sum of binomial coefficients indexed by the prime divisors of n , which we'll just denote as $f(n)$.

Introduction

For a positive integer n , define

$$f(n) = \sum_{p|n} \binom{n}{p}$$

where the sum is over the prime divisors p of n . For example

$$\begin{aligned} f(6) &= \binom{6}{2} + \binom{6}{3} \\ &= 15 + 20 \\ &= 35 \end{aligned}$$

and

$$\begin{aligned} f(30) &= \binom{30}{2} + \binom{30}{3} + \binom{30}{5} \\ &= 435 + 4060 + 142506 \\ &= 147001 \end{aligned}$$

It is interesting to note immediately that $f(n)$, for the lack of a better wording, seems to remember the numbers n/p . In particular, if q is a prime divisor of n , then

$$f(n) \equiv \frac{n}{q} \pmod{q}$$

This gives a surprising bridges between $f(n)$ and squarefree numbers, prime powers, perfect numbers, Giuga numbers, primary pseudoperfect numbers, a conjecture involving Carmichael numbers etc.

Elementary Results Involving $f(n)$

This section contains some elementary identities involving $f(n)$ alone. Some are just direct consequences of the definition, but the congruence result below seems to be the main tool. For every prime divisor q of n we have

$$f(n) \equiv \frac{n}{q} \pmod{q}$$

Proof. We start with the following elementary identity

$$\binom{n}{p} = \frac{n}{p} \binom{n-1}{p-1}$$

which is valid whenever p is a prime divisor of n . Now consider the sum

$$f(n) = \sum_{p|n} \binom{n}{p}$$

and fix a prime divisor q of n .

First suppose $p \neq q$. Since both p and q are prime, dividing n by p does not remove the factor q . Hence

$$q \mid \frac{n}{p}$$

Therefore

$$\binom{n}{p} = \frac{n}{p} \binom{n-1}{p-1} \equiv 0 \pmod{q}$$

Now consider the term where $p = q$. We have

$$\binom{n}{q} = \frac{n}{q} \binom{n-1}{q-1}$$

Since $q \mid n$, we get

$$n-1 \equiv -1 \pmod{q}$$

$$n-2 \equiv -2 \pmod{q}$$

...

$$n-(q-1) \equiv -(q-1) \pmod{q}$$

Therefore

$$\begin{aligned} \binom{n-1}{q-1} &= \frac{(n-1)(n-2)\dots(n-q+1)}{(q-1)!} \\ &\equiv \frac{(-1)(-2)\dots(-(q-1))}{(q-1)!} \pmod{q} \end{aligned}$$

The numerator is

$$(-1)^{q-1}(q-1)!$$

If q is odd then $q-1$ is even, and if $q=2$ then $-1 \equiv 1 \pmod{2}$. So in every case

$$\binom{n-1}{q-1} \equiv 1 \pmod{q}$$

and hence

$$\binom{n}{q} \equiv \frac{n}{q} \pmod{q}$$

All other terms in the sum are 0 modulo q , so

$$f(n) \equiv \frac{n}{q} \pmod{q}$$

□

For every positive integer n ,

$$n \text{ is squarefree} \iff \gcd(n, f(n)) = 1$$

Proof. Let q be a prime divisor of n . From the previous proposition

$$f(n) \equiv \frac{n}{q} \pmod{q}$$

Therefore

$$q \mid f(n) \iff q \mid \frac{n}{q}$$

which is equivalent to

$$q^2 \mid n$$

So a prime divisor q of n also divides $f(n)$ exactly when q^2 divides n . Therefore n has no repeated prime factor if and only if no prime divisor of n divides $f(n)$, which is exactly

$$\gcd(n, f(n)) = 1$$

□

For $n > 1$ we have

$$n \text{ is prime} \iff f(n) = 1$$

Proof. If n is prime, then the only prime divisor of n is n itself. Therefore

$$f(n) = \binom{n}{n} = 1$$

Conversely, suppose n is composite. Then every prime divisor p of n satisfies $p < n$, and hence

$$\binom{n}{p} > 1$$

Since $f(n)$ contains at least one such positive term, we get

$$f(n) > 1$$

Therefore $f(n) = 1$ can happen only when n is prime.

□

For every prime p and every integer $a \geq 1$,

$$p^{a-1} \mid f(p^a)$$

but

$$p^a \nmid f(p^a)$$

Proof. If $n = p^a$, then the only prime divisor of n is p , and so

$$f(p^a) = \binom{p^a}{p}$$

Using the binomial identity from above we get

$$\begin{aligned} f(p^a) &= \frac{p^a}{p} \binom{p^a - 1}{p - 1} \\ &= p^{a-1} \binom{p^a - 1}{p - 1} \end{aligned}$$

Thus

$$p^{a-1} \mid f(p^a)$$

It remains to show that the second factor is not divisible by p . Since $p \mid p^a$, we have

$$\begin{aligned} p^a - 1 &\equiv -1 \pmod{p} \\ p^a - 2 &\equiv -2 \pmod{p} \\ &\dots \\ p^a - (p - 1) &\equiv -(p - 1) \pmod{p} \end{aligned}$$

So

$$\binom{p^a - 1}{p - 1} \equiv 1 \pmod{p}$$

and hence this factor is not divisible by p . Therefore $f(p^a)$ is divisible by p^{a-1} but not by p^a . $\square$

Powers of two and perfect numbers

Consider $n = 2^a$. Then the only prime divisor of n is 2, so

$$\begin{aligned} f(2^a) &= \binom{2^a}{2} \\ &= \frac{2^a(2^a - 1)}{2} \\ &= 2^{a-1}(2^a - 1) \end{aligned}$$

This is the same expression that appears in the classical formula for even perfect numbers. For example

$$f(2^2) = f(4) = 6$$

and 6 is perfect. Also

$$f(2^3) = f(8) = 28$$

and 28 is perfect. Also

$$f(2^5) = f(32) = 496$$

and 496 is perfect.

The Euclid-Euler theorem says that the even perfect numbers are exactly the numbers of the form

$$2^{p-1}(2^p - 1)$$

where p is prime and $2^p - 1$ is also prime. Therefore the special values $f(2^a)$ are connected to Mersenne primes by

$$f(2^a) \text{ is perfect} \iff 2^a - 1 \text{ is prime}$$

for $a > 1$. This is a rather simple observation, but it is interesting that the binomial sum naturally produces the even perfect number expression.

A simple squarefree congruence

Now define

$$S(n) = \sum_{p|n} \frac{n}{p}$$

This is always an integer. If n is squarefree, then $f(n)$ and $S(n)$ are congruent modulo every prime divisor of n . If n is squarefree, then

$$f(n) \equiv S(n) \pmod{n}$$

Proof. Let q be a prime divisor of n . From the main congruence

$$f(n) \equiv \frac{n}{q} \pmod{q}$$

Now reduce $S(n)$ modulo q . We have

$$S(n) = \frac{n}{q} + \sum_{p|n, p \neq q} \frac{n}{p}$$

If $p \neq q$, then $q \mid n/p$, so

$$\frac{n}{p} \equiv 0 \pmod{q}$$

Therefore

$$S(n) \equiv \frac{n}{q} \pmod{q}$$

Hence

$$f(n) \equiv S(n) \pmod{q}$$

for every prime divisor q of n . Since n is squarefree, the prime divisors of n are distinct and their product is n . Therefore the congruences combine to give

$$f(n) \equiv S(n) \pmod{n}$$

□

This is the main bridge between $f(n)$ and sums of reciprocals of prime divisors, because

$$S(n) = n \sum_{p|n} \frac{1}{p}$$

Relation to Giuga numbers

A composite number n is called a Giuga number if

$$p \mid \frac{n}{p} - 1$$

for every prime divisor p of n . Equivalently

$$\frac{n}{p} \equiv 1 \pmod{p}$$

for every prime divisor p of n . For $n > 1$,

$$f(n) \equiv 1 \pmod{n}$$

if and only if n is either prime or a Giuga number.

Proof. First suppose n is prime. Then from above

$$f(n) = 1$$

and so

$$f(n) \equiv 1 \pmod{n}$$

Now suppose n is composite and

$$f(n) \equiv 1 \pmod{n}$$

Let q be a prime divisor of n . Then

$$f(n) \equiv 1 \pmod{q}$$

But by the main congruence

$$f(n) \equiv \frac{n}{q} \pmod{q}$$

Therefore

$$\frac{n}{q} \equiv 1 \pmod{q}$$

which is exactly the Giuga condition.

Conversely, suppose n is a Giuga number. Then for every prime divisor q of n ,

$$\frac{n}{q} \equiv 1 \pmod{q}$$

By the main congruence

$$f(n) \equiv 1 \pmod{q}$$

The Giuga condition also implies that $q^2 \nmid n$ for every $q \mid n$, because if $q^2 \mid n$, then

$$\frac{n}{q} \equiv 0 \pmod{q}$$

not 1 modulo q . Therefore n is squarefree. Since $f(n) \equiv 1 \pmod{q}$ for every prime divisor q of the squarefree number n , we get

$$f(n) \equiv 1 \pmod{n}$$

□

The number 30 is a Giuga number, and

$$f(30) \equiv 1 \pmod{30}$$

Proof. The prime divisors of 30 are 2, 3, 5. We have

$$\frac{30}{2} = 15 \equiv 1 \pmod{2}$$

$$\frac{30}{3} = 10 \equiv 1 \pmod{3}$$

$$\frac{30}{5} = 6 \equiv 1 \pmod{5}$$

Therefore 30 is a Giuga number. Also

$$f(30) = 147001$$

and

$$147001 \equiv 1 \pmod{30}$$

as expected.

□

Relation to primary pseudoperfect numbers

A positive integer n is called primary pseudoperfect if

$$\frac{1}{n} + \sum_{p|n} \frac{1}{p} = 1$$

Multiplying through by n gives

$$1 + \sum_{p|n} \frac{n}{p} = n$$

and therefore

$$1 + S(n) = n$$

So

$$S(n) \equiv -1 \pmod{n}$$

This gives another simple congruence for $f(n)$. If n is primary pseudoperfect, then

$$f(n) \equiv -1 \pmod{n}$$

Proof. First, a primary pseudoperfect number must be squarefree. Suppose $q^2 \mid n$ for some prime q . From

$$1 + \sum_{p|n} \frac{n}{p} = n$$

reduce both sides modulo q . The right side is 0 modulo q . Also, every term n/p is divisible by q . If $p = q$, this is because $q^2 \mid n$, and if $p \neq q$, this is because dividing by p does not remove the factor q . So the left hand side is congruent to

$$1 \pmod{q}$$

which is impossible. Therefore n is squarefree.

Since n is squarefree, we know that

$$f(n) \equiv S(n) \pmod{n}$$

But for a primary pseudoperfect number

$$1 + S(n) = n$$

and hence

$$S(n) \equiv -1 \pmod{n}$$

Therefore

$$f(n) \equiv -1 \pmod{n}$$

□

For example, 6 is primary pseudoperfect since

$$\frac{1}{6} + \frac{1}{2} + \frac{1}{3} = 1$$

and

$$f(6) = 35 \equiv -1 \pmod{6}$$

Semiprimes

Let

$$n = pq$$

where p and q are distinct primes. Then

$$f(pq) = \binom{pq}{p} + \binom{pq}{q}$$

By the main congruence

$$f(pq) \equiv q \pmod{p}$$

and

$$f(pq) \equiv p \pmod{q}$$

But the number $p + q$ satisfies the same two congruences

$$p + q \equiv q \pmod{p}$$

and

$$p + q \equiv p \pmod{q}$$

Therefore

$$f(pq) \equiv p + q \pmod{pq}$$

Thus, for a product of two distinct primes, the value

$$f(n) \pmod{n}$$

remembers the sum of the two prime factors. Since $p + q < pq$ for distinct primes p and q , this actually remembers the exact value of $p + q$. If $n = pq$ for distinct primes p and q , then

$$f(n) \equiv p + q \pmod{n}$$

This result is interesting because knowing $n = pq$ and $p + q$ gives the quadratic

$$x^2 - (p + q)x + n = 0$$

whose roots are p and q . So in the semiprime case, the residue of $f(n)$ modulo n contains the factor-sum information.

Generating functions

The ordinary generating function of $f(n)$ can be written directly from the definition. We have

$$\sum_{n \geq 1} f(n)x^n = \sum_p \sum_{m \geq 1} \binom{pm}{p} x^{pm}$$

This is because the condition $p \mid n$ means that

$$n = pm$$

for some integer $m \geq 1$.

There is also a simple exponential generating function. Starting from the definition

$$\sum_{n \geq 1} f(n) \frac{x^n}{n!} = \sum_p \sum_{m \geq 1} \binom{pm}{p} \frac{x^{pm}}{(pm)!}$$

Using

$$\binom{pm}{p} = \frac{(pm)!}{p!(pm-p)!}$$

we get

$$\begin{aligned} \sum_{n \geq 1} f(n) \frac{x^n}{n!} &= \sum_p \sum_{m \geq 1} \frac{x^{pm}}{p!(pm-p)!} \\ &= \sum_p \frac{x^p}{p!} \sum_{m \geq 1} \frac{x^{p(m-1)}}{(p(m-1))!} \end{aligned}$$

So

$$\sum_{n \geq 1} f(n) \frac{x^n}{n!} = \sum_p \frac{x^p}{p!} \sum_{m \geq 1} \frac{x^{p(m-1)}}{(p(m-1))!}$$

For example, the contribution from the prime 2 is

$$\sum_{m \geq 1} \binom{2m}{2} x^{2m}$$

Since

$$\binom{2m}{2} = m(2m-1)$$

we get

$$\sum_{m \geq 1} \binom{2m}{2} t^m = \sum_{m \geq 1} m(2m-1)t^m$$

Using

$$\sum_{m \geq 1} mt^m = \frac{t}{(1-t)^2}$$

and

$$\sum_{m \geq 1} m^2 t^m = \frac{t(1+t)}{(1-t)^3}$$

we obtain

$$\begin{aligned} \sum_{m \geq 1} m(2m-1)t^m &= 2 \sum_{m \geq 1} m^2 t^m - \sum_{m \geq 1} m t^m \\ &= \frac{2t(1+t)}{(1-t)^3} - \frac{t}{(1-t)^2} \\ &= \frac{t(1+3t)}{(1-t)^3} \end{aligned}$$

Therefore

$$\sum_{m \geq 1} \binom{2m}{2} x^{2m} = \frac{x^2(1+3x^2)}{(1-x^2)^3}$$

Relation to Carmichael numbers

A composite number n is called a Carmichael number if

$$a^{n-1} \equiv 1 \pmod{n}$$

for every integer a such that

$$\gcd(a, n) = 1$$

A standard elementary test for Carmichael numbers is Korselt's criterion, which states that a composite integer n is Carmichael if and only if n is squarefree and

$$p-1 \mid n-1$$

for every prime divisor p of n .

There is a known conjecture saying that no composite number is both a Giuga number and a Carmichael number. Using the $f(n)$ function, the Giuga condition can be written as

$$f(n) \equiv 1 \pmod{n}$$

and the Carmichael condition can be written as

$$p-1 \mid n-1$$

for every prime divisor p of n , together with squarefreeness. Since

$$f(n) \equiv 1 \pmod{n}$$

already forces squarefreeness for composite n , the conjecture can be restated in terms of $f(n)$ as follows. There is no composite integer n such that

$$f(n) \equiv 1 \pmod{n}$$

and

$$p - 1 \mid n - 1$$

for every prime divisor p of n . This formulation is interesting because it combines two different-looking conditions.